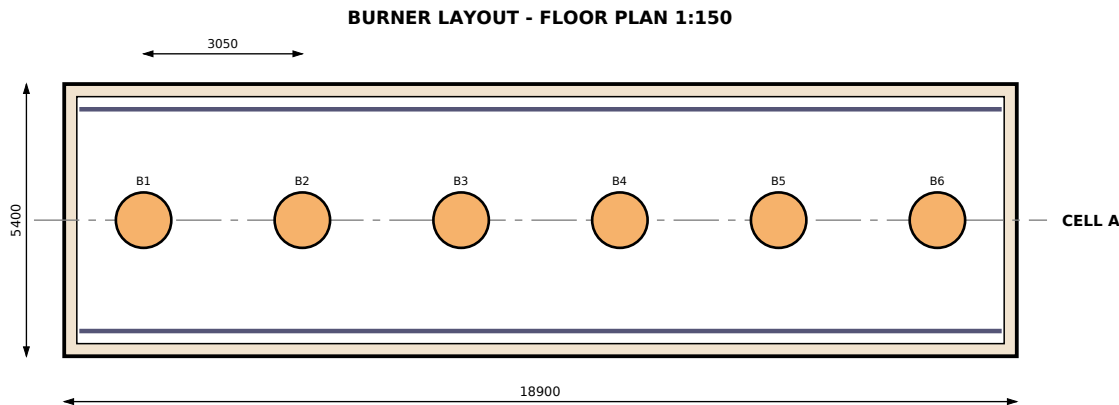
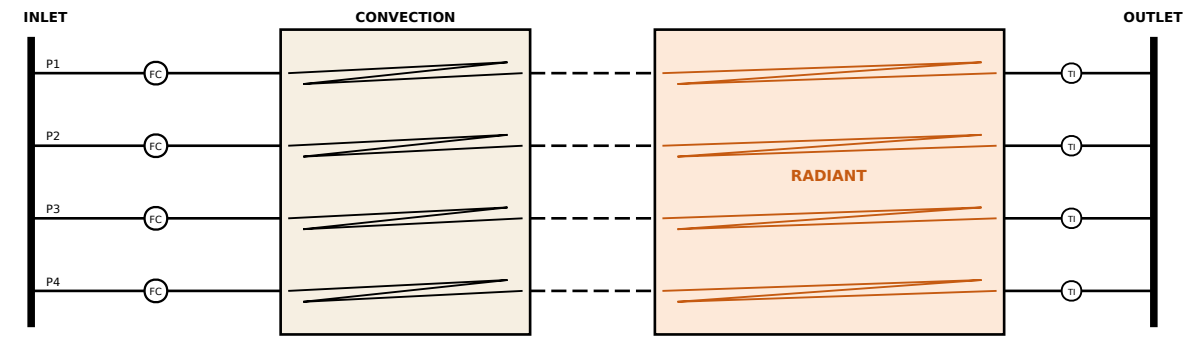


H-201 COIL SCHEDULE								
COIL SECTION	ROWS	TUBES/ROW	No. TUBES	OD x WALL mm	MATERIAL	EXT. SURFACE	DUTY MW	TDM C
Radiant	22 wall	-	44	168.3 x 10.97	A335 P9	bare	10.3	575
Convection - shock	2	4	8	168.3 x 7.11	A335 P5	bare	-	-
Convection - studded	3	4	12	168.3 x 7.11	A106 B / P5	studs 12.7x25	4.4	-
Utility coil (HOLD)	2	4	8	114.3 x 6.02	TBD	TBD	0.77	-



PASS ARRANGEMENT (NTS)



4 passes x 64.5 t/h; pass flow control + pass outlet TI (balancing), COT to C-201 transfer line

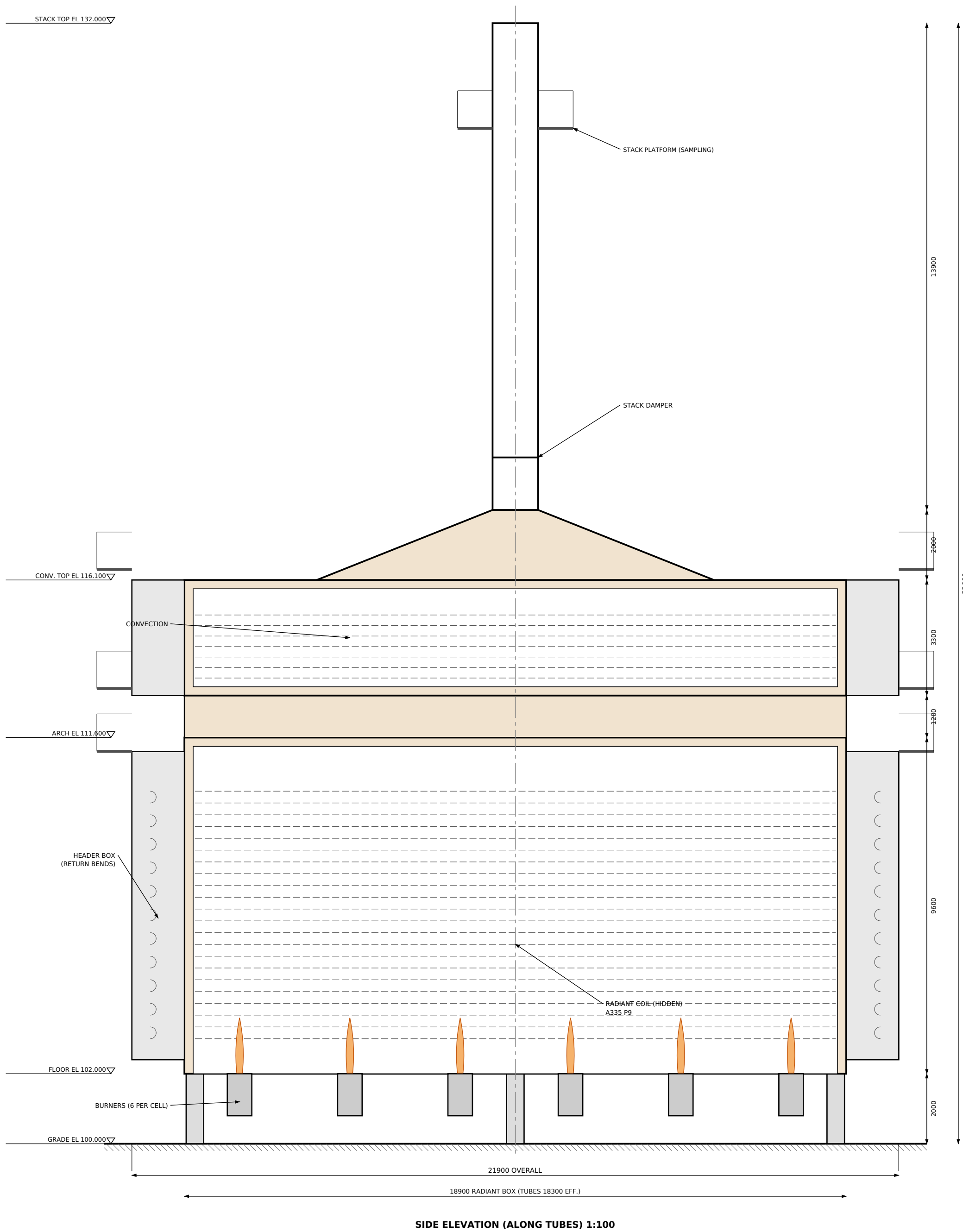
HOLD - HEAT RECOVERY

Process convection can recover only 3.6 MW (flue >= 399 C).
0.8 MW utility coil shown dotted (steam superheat / BFW) - or add APH.
Off-gas (stream 28, from D-202) burned on dedicated burner tips.

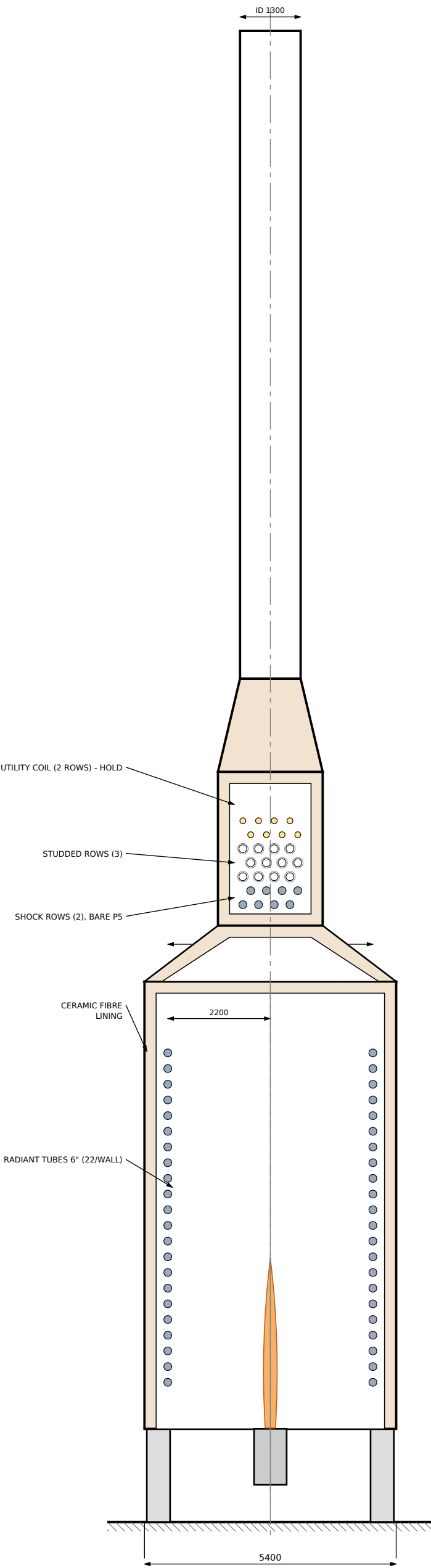
H-201 DESIGN DATA	
Service	Vacuum heater
Type	1-cell cabin, horizontal tubes, natural draft
Duty absorbed / fired	14.7 / 17.5 MW (LHV)
Radiant / convection duty	10.3 / 4.4 MW
Efficiency (LHV)	84 %
Flow / passes	258 t/h / 4
Inlet / outlet T	349 / 398 C
Outlet vaporisation	26 wt %
Avg. radiant flux	25.0 kW/m2
Mass flux	1066 kg/m2s
Bridgwall T (est.)	822 C
Flue gas	25 t/h
Overall L x W x H	21.9 x 5.4 x 32 m
Weight empty / operating	340 / 349 t

COIL DATA (API 530 - CFU-000-ME-CAL-001 S10)	
Radiant tubes	44 x 168.3 OD, Sch 80 (10.97 mm), A335 P9
Effective length	18.3 m, 22 per wall per cell
Design P elastic / rupture	19.8 / 19.0 barg
Design metal T (API 530)	575 C (EOR TMT 559 C)
1 min elastic / rupture	5.11 / 6.25 mm (rupture governs)
Corr. allow. / life	3.0 mm / 100,000 h
Convection	20 tubes: 2 shock + 3 studded rows x 4
Extra coils	utility 2 rows (HOLD, 0.8 MW)
Tube supports	HK40 radiant / 50Cr-50Ni-Nb shock

BURNERS (API 535)	
Burners	6 ULNB staged-fuel gas, up-fired
Arrangement	single row per cell, pitch 3.05 m
Heat release design / max	2.92 / 3.64 MW
Burner CL to tube CL	2.20 m
Pilots / scanners	gas pilot, HEL, UV scanner each



SIDE ELEVATION (ALONG TUBES) 1:100



SECTION A-A (ACROSS CELLS) 1:100

NOTES

- Dimensions in mm, elevations in m; grade EL 100,000.
- FEED arrangement - vendor to optimise (API 560).
- Radiant floor 2.0 m above grade for burner access.
- Tube pulling: removable header-box doors at both ends; crane access required.
- Datasheet CFU-000-ME-DS-003; coil design CFU-000-ME-CAL-001 S10.
- HOLD: 88 % efficiency needs - 0.8 MW utility coil or APH (CAL-001 S13).
- HOLD: plot L 21.9 m in equipment, json < 21.9 m required.
- Outlet tube step-up (6"->8"->10") for vacuum service by vendor.

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
FIRED HEATER GENERAL ARRANGEMENT H-201 VACUUM HEATER			
DISCIPLINE		STAGE	SHEET
MECHANICAL		FEED	1 OF 1
DRAWN		CHECKED	DATE
CLAUDE CODE		-	2026-10-02
DRAWING NUMBER			REV
CFU-200-ME-GA-006			A